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Investment Strategies

Expected Skewness and Momentum

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Introduction

The aim of this paper is to analyze and empirically replicate, through implementation in MATLAB, the strategy proposed in the paper *"Expected Skewness and Momentum"* written by *Jacobs H., Regele T. and Weber M*. The interest of the study lies in the possibility of improving the robustness and stability of traditional momentum strategies, often subject to phases of negative returns concentrated in periods of strong market correction, by integrating a measure of yield asymmetry, i.e. *expected skewness*, into the security selection process. The paper shows that conventional momentum strategies suffer substantial losses during market reversals due to the negative *momentum skewness*, linked to the strong rebounds of previously losing stocks. The authors link this phenomenon to the asymmetry of individual stock returns and propose to evaluate it through *expected skewness* as a tool to mitigate such crash phases. This measure is calculated, within the replica, as the maximum daily logarithmic return of the previous month and serves as an ex-ante indicator of the propensity of securities to generate asymmetric and potentially unstable returns. By combining this variable with the *momentum* signal, the authors construct a double sorting dependent on quintiles, this procedure is recalibrated on a monthly basis, positions are closed and reopened at the end and beginning of each month, in order to update the composition of the portfolios according to the joint changes in *momentum* and *skewness*. Three main portfolios are derived from this methodology:

- *Standard*: acts as a benchmark, within the third skewness quintile, a long vs short portfolio is constructed, with stock selection based on *momentum*.
- *Weakened*: assumes positive exposures for stocks with high expected skewness and high momentum, negative exposures for stocks with *low skewness* and *low momentum*.
- *Enhanced*: takes bullish positions for stocks within the top skewness quintile that have high momentum and bearish positions for stocks in the bottom skewness quintile with *low momentum*.

In addition, there is the *Momentum Neutral* portfolio, defined as the difference between the *Enhanced* and *Weakened* portfolios, built to measure the net effect of *expected skewness* on the profitability of the *momentum* factor, showing the extent to which selection based on asymmetry of returns contributes to improving the traditional strategy. In the original work, two dynamic risk management constraints are also introduced: the *Barroso and Santa Clara* rule, based on *volatility targeting*, and the rule devised by *Daniel* and *Moskowitz*, which reduces the strategy's exposure to negative and highly volatile market states. Both are intended to mitigate the variability of returns and improve *risk-adjusted* performance. The replication presented in this study adopts, in line with the methodology of the original paper, a *zero-invested* logic, in which the sum of the exposures deriving from the long positions and that of the short positions are of equal size. However, it is based on data referring to a more recent and shorter time period (2006–2024), this choice reflects the intent to verify the temporal robustness of the evidence found by the original study, observing whether the relationships between *skewness, momentum* and *tail risk* persist even in a modern market context, characterized by episodes of concentrated volatility (2008 crisis, 2020 pandemic, recent geopolitical shocks) and greater information efficiency. With the adoption of risk management rules and a framework that estimates cumulative returns, performance metrics, volatility and multi-factor Fama–French regressions, the goal is to verify whether, on a contemporary sample, the *Skewness-Enhanced Momentum portfolio* maintains, as in the original experiment, higher and more stable returns than traditional *momentum* and whether the *risk-managed* versions increase their resilience to negative tails.

Dataset construction

The first phase necessary for replication was the construction of a dataset consistent with the original strategy, as previously mentioned the experiment was conducted considering only the time period from 2006 to 2024. Using the Factset software, a list of 300 titles was generated through the screening function. These assets were selected by adding several filters, which allowed only stocks with these characteristics to be considered:

- belonging exclusively to the North American market, to ensure geographical consistency with the original composition of the sample;
- classification as ordinary shares, thus excluding preferential securities, ADRs or non-comparable hybrid instruments;
- price above $5 on January 1, 2006, in order to exclude penny stocks and reduce potential distortions due to low liquidity;
- market capitalization of more than $5 billion on January 1, 2006, in order to keep only large companies in the sample;
- initially listed on the US regulated exchanges NYSE and NASDAQ, in order to ensure the inclusion of the most representative companies of the US stock market of the US stock market.

After selection, the historical prices of the securities were downloaded in *an adjusted* format, including dividends and extraordinary transactions, such as stock splits and mergers, in order to obtain comparable effective returns over time. The data was then cleaned by eliminating outliers or unavailable values and aligned by date, so as to ensure a consistent and uninterrupted time series.
The dataset construction logic has been designed to minimize *survivorship bias*, ensuring a more realistic representation of the market. To this end, delisted or bankrupt securities have also been included, so as not to limit the analysis to only the assets that survived in the observation period. This avoids an artificial overestimation of historical performance, as it also takes into account the negative returns associated with companies that have ceased to be listed.

The basis thus constructed was the input for the calculation of daily and monthly logarithmic returns using MATLAB and for the subsequent implementation of the investment strategy.

Performance and volatility metrics

Within this report, in order to compare the different strategies on which the backtest was carried out, different measures of performance and volatility are presented, which are calculated over the entire analysis period and presented later.

Performance Measures

Performance metrics are used to assess the profitability of strategies and the risk related to each strategy. These measures allow the quality of returns to be analysed in relation to risk and include both absolute and risk-adjusted measures. Because strategy comparisons are only meaningful on a consistent basis, all key performance metrics are annualized, i.e., reported on an annual basis, even if calculated over the entire backtest period. This allows them to be interpreted as annual average values and to compare them directly between strategies with different durations or with market benchmarks:

- *CAGR (Compound Annual Growth Rate):* represents the compound annual growth rate that an investment would have had if it had grown at a constant rate each year, starting from an initial value to a final value, in the period considered. It provides a measure of average annual return, independent of intermediate fluctuations. A high CAGR indicates that the investment has grown steadily and steadily over time, while a low value suggests modest or negative growth over the long term.
- *Sharpe Ratio*: measures risk-adjusted return, comparing the average excess return against a risk-free rate (set at 0 in our analysis) to the annualized volatility of returns. A high value indicates that the strategy has generated a significant return for each unit of risk taken, while low values indicate insufficient returns relative to the risk incurred. In general, a Sharpe Ratio greater than 1 is considered positive, while values below 1 indicate an insufficient return compared to the risk.
- *Sortino Ratio*: it is a variant of Sharpe that considers only downside risk, measuring the volatility of negative returns compared to the average return. Again, the measure is annualized to ensure consistency with the other metrics, a high Sortino Ratio indicates that the strategy has achieved good returns while limiting losses.
- *Max Drawdown*: Measures the maximum loss incurred by a portfolio compared to its all-time high during the *backtest* period. It is a purely cumulative measure and is not annualized, as it represents an extreme event observed over the entire time horizon, lower values indicate greater resilience of the portfolio in times of market stress.

Volatility Measures

Volatility measures describe the risk profile of strategies, indicating how far returns deviate from their average and how the tails of the distribution behave. Unlike performance metrics, not all volatility measures are annualized: some describe distributional properties of the return series and must be interpreted on the original frequency of the data (in this case monthly):

- *Volatility:* represents the standard deviation of returns and measures the overall variability of the strategy, it is also used in the calculation of indicators such as Sharpe and Sortino, it is therefore calculated on an annual basis, to ensure consistency and allow direct comparisons between strategies and with market benchmarks.
- *Skewness: Measures* the skewness of the yield distribution, indicating whether the tails of the distribution are concentrated towards positive or negative values. As this is an intrinsic property of the distribution, it is calculated over the entire backtest period and reflects the overall shape of the distribution of returns.
- *Percentile at 1% (Q01):* represents the minimum return observed in the worst 1% of cases, indicating tail risk and the probability of extreme losses. This measure is also distributional in nature and is maintained over the monthly frequency of the data, as it describes the behavior of the distribution's queues as it manifests itself in the original data. Changing its time base would alter its meaning.

These performance and volatility measures, calculated consistently over the entire *backtest* period, provide a comprehensive overview of the returns and risks associated with each strategy. Comparing them allows for an in-depth analysis of the efficiency, stability and risk-return profile of different investment approaches.

Omega Metric

In addition to the calculation of the performance and volatility metrics necessary for the replication of the original strategy, in order to have a synthetic, universal and intuitive metric for comparing the different strategies and portfolios, a proprietary indicator called Omega Portfolio Rating (OPR) has been developed. The objective of the OPR is to express, in percentage terms, the improvement or worsening of a strategy compared to a reference benchmark – in this case, the *Equally Weighted* portfolio of the 300 stocks that make up the dataset – through a weighted combination of three independent measures of investment efficiency and robustness.

In particular, the OPR represents the average of the percentage change of three specific metrics, selected to offer an overall assessment of the strategy's *risk-adjusted* profile :

- *Calmar Ratio*, a consolidated measure that expresses the ratio between the *Compound Annual Growth Rate* (CAGR) and the maximum *drawdown*; it assesses the efficiency of the return adjusted for the maximum incurred loss.
- *Modified Sortino Ratio,* a variant of the traditional Sortino ratio, in which the denominator is not the standard deviation of negative returns, but the arithmetic mean of negative returns. This change emphasizes the penalty due to the frequency and intensity of average losses, rather than their variability.
- *Validity Index or Robust,* a new ad hoc indicator, designed to measure the solidity and consistency of performance over time. In the numerator, it considers the cumulative return of the strategy excluding the top 1% of returns, so as to reduce the impact of positive and isolated extreme events. The denominator is instead the average between the maximum duration (in terms of time) of *drawdowns* and the average duration of the *overall drawdowns* observed, thus providing a measure of risk in time dimension.

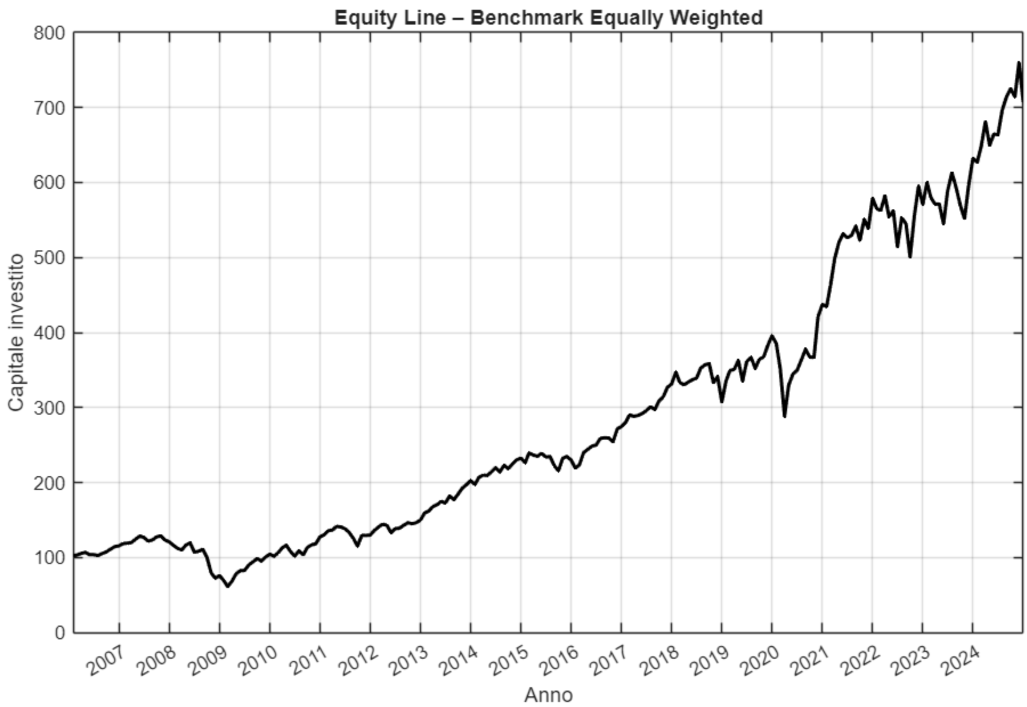
Taken together, these three components make it possible to assess in an integrated way the strategy's ability to generate sustainable and regular returns, while limiting exposure to prolonged losses.

The main purpose of using this statistic is to compare the portfolios defined by the paper with a passively managed strategy representative of the underlying market, in order to verify whether the relationship between *momentum* and *expected skewness*, in addition to maintaining internal validity, is actually able to generate a higher return than a diversified investment without active management.
The final result of the *Omega Portfolio Rating* therefore provides a synthetic index (centered on zero) that allows you to appreciate, in an immediate and comparable way,

how much a given strategy exceeds or underperforms the *benchmark* in terms of overall efficiency and resilience of the risk-return profile.

Since the main objective is the comparative evaluation of strategies and their combinations, quantitative analyses are interpreted from a diagnostic and not a predictive perspective. The purpose is to identify patterns of efficiency and dynamic behavior, not to build a trading model.

The wealth curve of the reference benchmark and its three metrics used for the calculation of the OPR metric for all portfolios are presented below:



The graph above represents the evolution of the equally *weighted* passive strategy built on the 300 stocks in the dataset, realistically considering the impact of delistings and bankruptcies. In the event of an exit from the market due to delisting, the price series is interrupted and the next return is considered to be zero, simulating a refund to the value of the last available quote. In this way, the benchmark accurately reflects the behavior of a constantly rebalanced passive portfolio, including the economic effects of any failures and ensuring comparability with active strategies in terms of long-term efficiency and resilience.

Strategy	**SortinoMod**	**Calmar**	**Robust**
BENCH	0,25137	0,20476	0,20145

The benchmark metrics show a solid but improvable risk-return profile. Calmar reflects the presence of significant *drawdowns* in the face of high growth, while the modified Sortino shows a good ability to generate returns despite non-negligible negative losses, finally, the Robust metric indicates a distribution of performance that is not perfectly regular but consistent with a diversified equity portfolio.

Overall, the benchmark combines a very high absolute return with a balanced risk-return profile.

Initial premises

In the initial phase of the strategy, using MATLAB, the daily logarithmic returns of each of the 300 stocks were extracted from the dataset in Excel format, which were then transformed on a monthly basis.

In this same phase, the case of delisting was also distinguished from that of actual bankruptcy: in the first case, a full repayment of the capital was assumed, assigning a return of zero at the time of removal from the list; in the second, corresponding to a corporate default, a total loss of 100% of the invested value was recorded.

Subsequently, using daily and monthly logarithmic returns as basic variables, the monthly *momentum* and *expected skewness values* for each security were calculated according to the following procedures:

- *Momentum*: estimated through a *twelve-month rolling window, calculating the cumulative logarithmic return of the previous eleven months and omitting the last month, in order to avoid* look-ahead bias. This specification corresponds to the conventional formulation of the *momentum literature*.
- *Expected skewness*: measured using daily data from the month prior to the holding period; in line with Bali et al, the proxy adopted corresponds to the maximum daily logarithmic return of the previous month, considered indicative of a stock's propensity to have positive tails in the distribution of returns.

The monthly values of the two metrics (*expected skewness* and *momentum*) were subsequently used for the *sorting* procedure, from which the construction of all the portfolios subject to the replication derives. In particular, for each month the distribution of *skewness* values is generated, ordered in ascending order, then divided into five quintiles. Each group thus obtained is treated as an independent subsample, within which the securities are further ordered, again with a classification by increasing values, based on their respective *momentum* values, also divided into five quintiles. This conditional dual ordering procedure makes it possible to analyse the combined effect of *skewness* and *momentum* on the profitability of portfolios, faithfully replicating the methodology adopted by the original strategy.

Development of basic strategies

The construction of the four main strategies represents the starting point for the empirical analysis of the relationship between *momentum* and *expected skewness*.
Each portfolio is obtained by specifically combining the quintiles of the two indicators, to isolate the different ways in which past performance and expected asymmetry of securities interact.

First, for each month, the equity universe is sorted by the distribution of *expected skewness* values and divided into five quintiles. Within each *skewness* quintile, stocks are further classified into five sub-quintiles according to their *momentum* value. This dual ordering makes it possible to construct *long–short zero-invested portfolios*, rebalanced monthly, which distinctly reflect the different combinations of the two dimensions of interest.

The four resulting strategies are defined as follows:

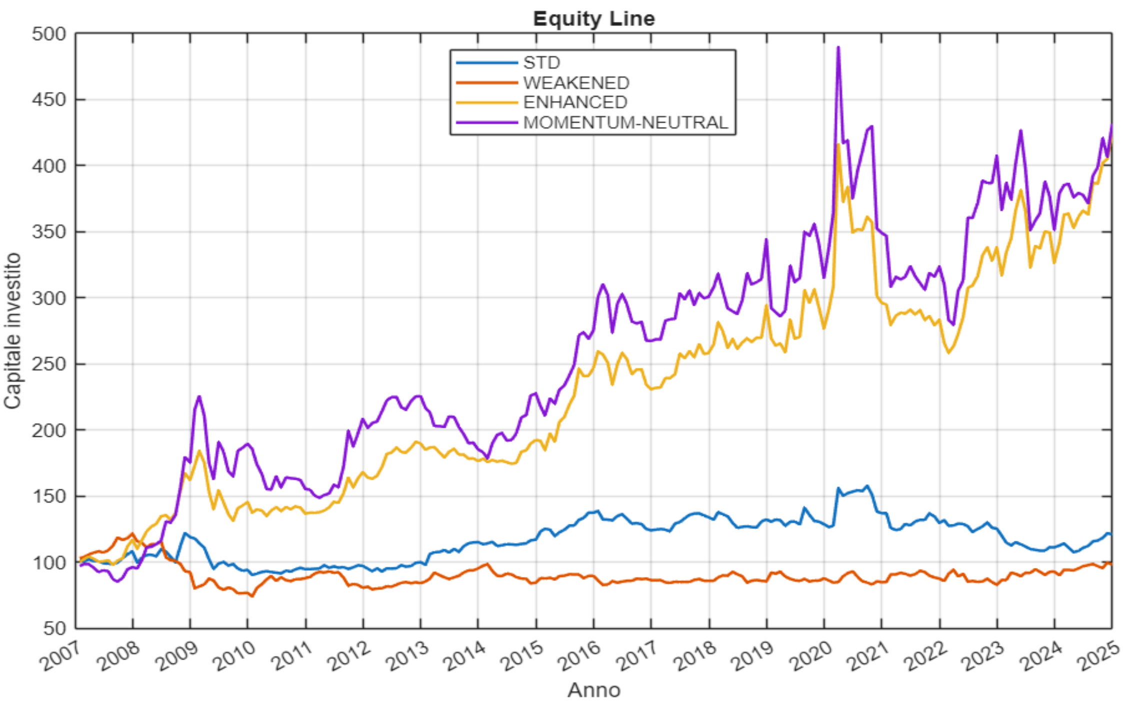
- **Standard Portfolio (Regular Momentum):**
  considers only the securities belonging to the central quintile of *skewness*, within which a long position is opened on the securities with higher *momentum* (fifth quintile) and a short position on those with lower *momentum* (first quintile).
- **Enhanced Momentum portfolio:**
  favors stocks with low *expected skewness* and high *momentum*, taking long positions in the fifth momentum quintile of the first *skewness quintile* and short positions in the first momentum quintile of the last *skewness* quintile. This combination is designed to maximise exposure to securities with strong past returns but less asymmetric distributions, reflecting the assumption that stable positive returns are associated with lower risks of significant crashes.
- **Weakened Momentum Portfolio:**
  follows the opposite logic, investing in stocks with high *expected skewness* and high *momentum* (long positions in the fifth momentum quintile of the last skewness quintile) and short those with low *expected skewness* and low *momentum* (first momentum quintile of the first quintile of *skewness*). This setup intentionally reduces the robustness of the momentum signal (it selects stocks with unstable trends), thus testing its sensitivity to the presence of positive tails in returns.
- **Momentum-Neutral Portfolio:**
  it is constructed as the difference between the monthly returns of the Enhanced and Weakened portfolios, and allows to eliminate the effect of the *momentum*

dimension on the return evaluated by the *skewness* metric, providing a synthetic indicator of the asymmetry component not explained by the pure dynamics of past returns.

The basic idea is that the relationship between *momentum* and *expected skewness* reflects not only the strength of the past trend, but also the shape of the distribution of expected returns. In particular, the *Enhanced* and *Weakened strategies* make it possible to verify whether momentum is more profitable when associated with securities with less asymmetric return distributions or whether the presence of positive tails represents an additional risk signal for the investor.

Subsequently, the results of the replication of the strategies are presented, expressed both through the main performance metrics and through the wealth curves. It should be noted that, since these are *zero-invested* strategies, the wealth curve represents a purely illustrative simulation, aimed at highlighting the cumulative trend of returns over time rather than the evolution of invested capital.

The chart clearly shows how the interaction between *momentum* and *expected skewness* decisively influences the profitability of portfolios. The *Enhanced Momentum* strategy is significantly superior to *the Weakened* and *Standard* strategies, with steady growth and more than quadrupled capital, in line with the original paper.



The *Momentum Neutral* portfolio follows a similar but more volatile trend, confirming the importance of the *skewness component*. Standard or *Regular Momentum* and *Weakened Momentum* provide almost zero returns, highlighting large *drawdown* phases.

Overall, the image confirms that the yield premium derives mainly from the selection of bearish positions towards stocks with positive expected *skewness* and bullish positions towards stocks with negative *expected skewness*, consistent with the empirical results of the paper.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Vol	Skew	Q01
STD	0,01068	0,15074	0,25581	0,31844	0,10741	1,8663	-0,08005
WEAKENED	-0,00098489	0,038782	0,051747	0,39072	0,097075	-0,55022	-0,080924
ENHANCED	0,083583	0,56148	0,91236	0,37928	0,16769	1,3892	-0,11914
MN	0,084651	0,48758	0,79954	0,42942	0,21174	0,86316	-0,15714

Strategy	SortinoMod	Calmar	Robust	Omega
STD	0,066925	0,033538	-0,0028938	-0,86144
WEAKENED	0,0080424	-0,0025207	-0,0013418	-0,99566
ENHANCED	0,25178	0,22327	0,053578	-0,21055
MN	0,23944	0,19713	0,041106	-0,293553
BENCH	0,25137	0,20476	0,20145	0

The metrics in the first table confirm that the differentiator of the most effective strategies is the intentional construction of portfolios that exploit the expected asymmetry in a targeted manner. Enhanced combines winners with negative *skewness* and losers with positive *skewness*, thus maximizing the reward associated with these characteristics and achieving the best results in terms of return and risk-reward ratio. Momentum *neutral*, created to isolate the effect of *skewness* by reducing exposure to trends, demonstrates how this dimension maintains a strong explanatory power regardless of the *momentum* component. Both, however, show high volatility and deep *drawdowns*, a sign that the search for excess returns through *skewness* implies greater overall risk.

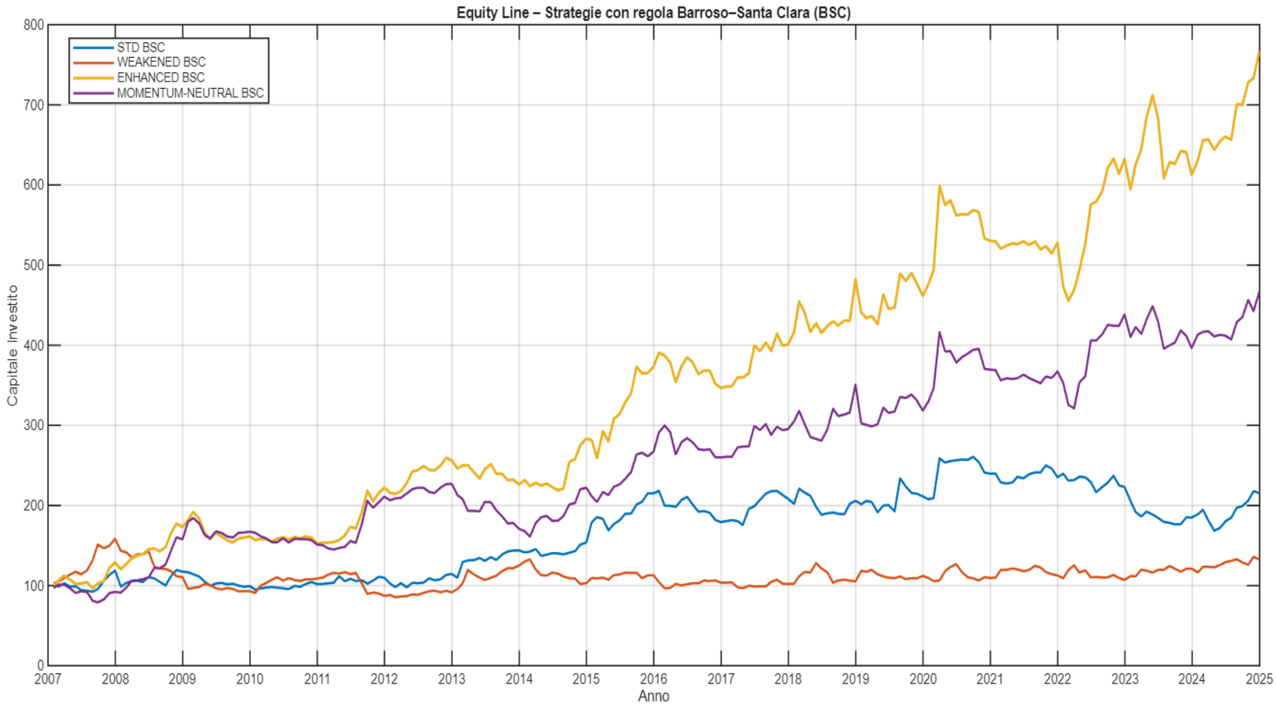
The analysis of the Omega metric, on the other hand, shows that, although strategies based on the interaction between *momentum* and *skewness* significantly improve their risk-return characteristics compared to less efficient configurations, none manages to exceed the benchmark, thus highlighting the limitations of these strategies without the implementation of *risk management rules*.

Implementation of the Barroso-Santa Clara rule:

The Barroso-Santa Clara rule has been implemented in the code as *dynamic volatility targeting*, scaling the portfolio's monthly exposure to keep annualized volatility close to a set target of 13%. The approach uses a rolling window of 12 months to estimate ex-ante volatility and applies an operational leverage limit between 0 and 3, with a one-month shift to avoid *look-ahead bias*.

$$r_t^{adj} = \frac{\sigma^*}{\hat{\sigma}_{t-1}} \cdot r_t$$

Compared to the original version of the model, which takes a window of 36 months, without defining any explicit constraint on leverage, this implementation is more responsive to regime changes, although it introduces greater variability in the leverage coefficient. Despite these differences, the fundamental rationale remains unchanged and empirical results show a substantial improvement in the risk-return profiles of strategies, in particular for the *Enhanced momentum portfolio*.



The implementation of this rule is clearly reflected in the evolution of equity lines, showing the effect of dynamic leverage in modulating exposure to different market regimes. Downsizing the position according to volatility achieves a more stable and gradual growth than the original strategies, with a reduction in the depth of *drawdowns* and greater continuity of returns over time.
Leverage increases exposure in periods of low volatility, amplifying the cumulative return especially for the *Enhanced* portfolio and, to a lesser extent, for *the Momentum-Neutral* portfolio, while it tends to reduce it in turbulent phases, helping to contain losses and preserve capital.
The result is an overall more regular capital dynamic, with stronger growth and less vulnerable to market shocks.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Vol	Skew	Q01
STD_BSC	0,04356	0,34355	0,63319	0,35354	0,16063	1,2326	-0,080071
WEAKENED_BSC	0,015835	0,17887	0,26216	0,46087	0,15327	0,083979	-0,13054
ENHANCED_BSC	0,11987	0,78566	1,5216	0,23964	0,16103	0,74231	-0,09864
MN_BSC	0,089432	0,60343	1,1063	0,29016	0,1642	0,76419	-0,10262

Strategy	SortinoMod	Calmar	Robust	Omega
STD_BSC	0,16782	0,12321	0,013166	-0,5551
WEAKENED_BSC	0,070501	0,04386	-0,0020379	-0,851
ENHANCED_BSC	0,39712	0,50021	0,24084	0,73942
MN_BSC	0,3179	0,30822	0,12404	0,12856
BENCH	0,25137	0,20476	0,20145	0

This change from the basic strategies, as can be seen from the first image, produces a systematic improvement in the risk-return profile compared to previous results. Firstly, a significant increase in *risk-adjusted performance ratios*, such as *Sharpe* and *Sortino*, is observed in all strategies, a sign that dynamic leverage can stabilize volatility and make returns more efficient. At the same time, the *Max Drawdown* decreases significantly, indicating a greater ability to contain losses in phases of market stress.
In terms of volatility, the strategy shows a more stable behavior: the overall level of risk is reduced, while the distribution of returns improves, with *Q01* less extreme and more balanced *skewness*.

The metrics used to calculate the Omega statistic also show clear progress: the *Modified Sortino* and the *Calmar* grow markedly, reflecting a better ratio between return and risk.

Overall, the application of the BSC rule transforms strategies that were originally more volatile and vulnerable into more stable, defensive portfolios capable of generating stronger returns over time.

Implementing the Daniel Moskowitz rule:

Daniel and Moskowitz's rule, created in 2016, was created to mitigate the so-called *momentum crashes*, i.e. the periods in which *momentum* strategies, normally profitable, suffer sudden and deep losses. These events usually occur after long market *drawdowns* or in phases of high volatility, when *reversals,* or reversals of market trends, become more likely.

To reduce risk, the authors propose to dynamically modulate the portfolio's exposure: it is reduced in dangerous regimes identified by two main signals:

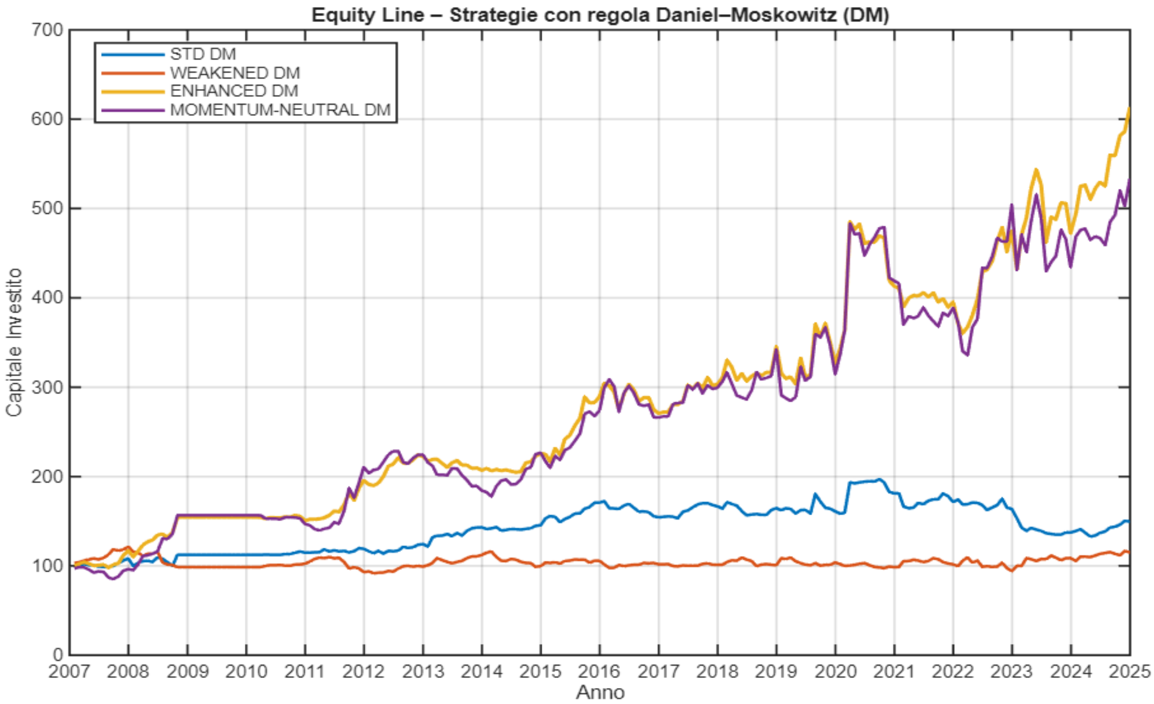
- Negative cumulative performance at 24 months
- volatility above the historical 75th percentile

This approach makes it possible to contain losses without altering the *momentum* logic, improving its risk-return profile and robustness in the long term.

The returns thus obtained, called *DM-scaled returns*, therefore represent the actual performance of the strategy after adjusting the risk exposure. They reflect a more conservative portfolio in unfavourable regimes and more aggressive in favorable ones.

$$r_t^{DM} = k_{DM,t} \cdot r_t$$

The implementation of the rule in the replication carried out through the code on MATLAB is consistent with the original strategy and as for the constraints introduced by the Barroso and Santa Clara methodology, it has brought improvements in terms of risk-return compared to the initial strategies.



As we can see, the application of the above-mentioned constraints significantly changes the dynamics of the strategies compared to their basic version, with more regular equity lines and more stable overall growth over time. In particular, the *Enhanced* and *Momentum neutral* strategies show a more gradual capital trajectory that is less prone to sudden swings.

Compared to the Barroso-Santa Clara rule, the trend is generally less volatile and characterized by more homogeneous phases of accumulation, albeit with an overall growth dynamic that appears to be more contained. These results suggest a significant impact of the exposure reduction component introduced by the DM rule, which translates, considering all four strategies, into a more controlled evolution of the portfolio.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Vol	Skew	Q01
STD_DM	0,022591	0,2685	0,44833	0,32648	0,10192	2,2062	-0,070733
WEAKENED_DM	0,0077874	0,13371	0,18166	0,24661	0,084924	-0,15606	-0,065681
ENHANCED_DM	0,10604	0,7299	1,2914	0,25701	0,15431	1,9663	-0,09453
MN_DM	0,097494	0,59714	0,9912	0,30479	0,18358	1,1289	-0,12905

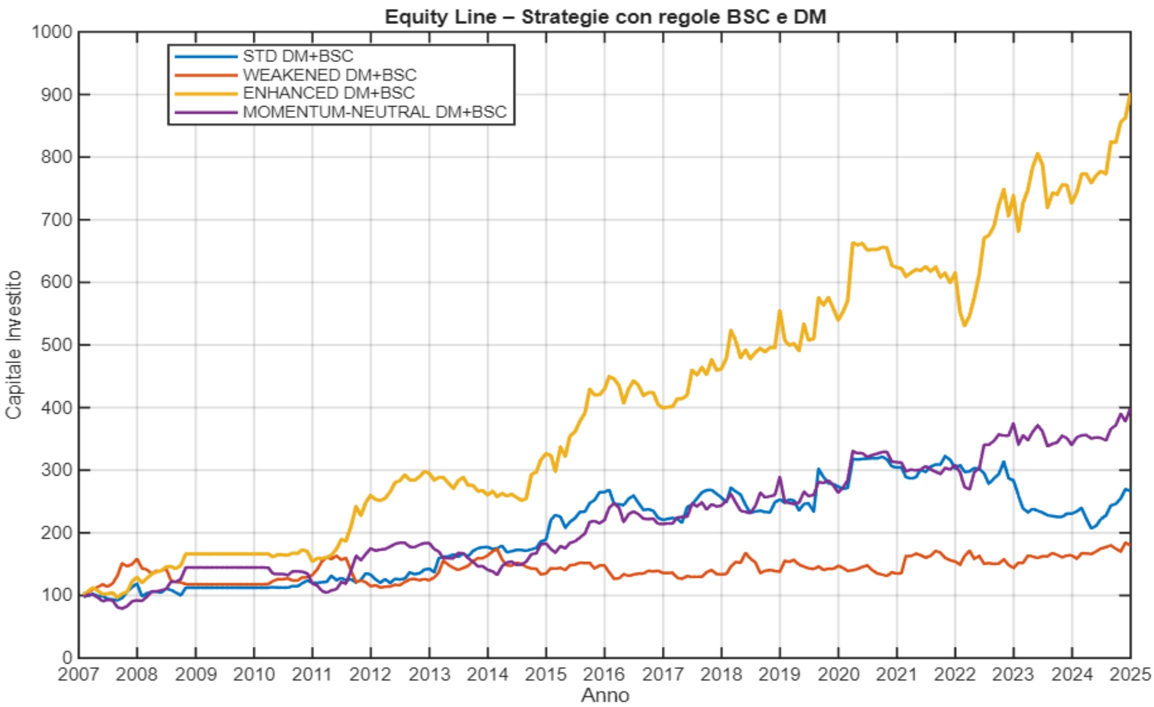
Strategy	SortinoMod	Calmar	Robust	Omega
STD_DM	0,12256	0,061997	0,020175	-0,72087
WEAKENED_DM	0,043802	0,031578	-0,00034882	-0,89109
ENHANCED_DM	0,36562	0,4126	0,13963	0,38755
MN_DM	0,29819	0,31987	0,12107	0,11648
BENCH	0,25137	0,20476	0,20145	0

By observing the first figure, we can understand how the use of this constraint produces a significant evolution of metrics compared to both basic strategies. First, performance and volatility metrics are noticeably better than the original model. In particular, there are higher results, also in this case, for *the Enhanced* and *Momentum neutral strategies*. Secondly, comparing these metrics of the strategies obtained through the DM rule with those obtained with the BSC rule, it emerges that the Daniel-Moskowitz approach does produce an improvement compared to the original model, but remains generally less effective than simple *volatility targeting*. The *CAGR, Sharpe* and *Sortino* values are in fact lower than those obtained with the Barroso-Santa Clara method, highlighting how intervention based on *drawdowns* and market volatility sometimes reduces exposure excessively, penalizing overall performance.
Overall, therefore, the implementation of the Daniel-Moskowitz methodology represents a step forward compared to the standard model, improving risk control and the stability of strategies, but does not achieve the effectiveness of the purely *volatility targeting* approach, which guarantees superior results in terms of performance and risk-return efficiency.

The comparison of Omega metrics fully confirms what has been observed in traditional performance measures. The strategies obtained through the DM methodology show values on average higher than the standard model, signaling an improvement in the balance between return, *drawdown* and robustness. However, the BSC approach continues to stand out as the most effective: its strategies achieve the highest values in all three components of Omega, reflecting more efficient risk management and a more favorable risk-reward ratio.
In summary, DM represents a step forward compared to the base, but BSC confirms itself as the most performing and robust solution even according to this composite metric.

Implementing strategies with combined rules:

This section presents the results obtained from the joint implementation of the two risk management rules, that of Barroso and Santa Clara, based on *volatility targeting* and that of Daniel and Moskowitz, based on the analysis of market regimes through cumulative *drawdowns* and volatility.
Unlike the original model proposed in the paper, which analyses the effectiveness of each approach separately, the methodology adopted here provides for a sequential combination of the two strategies, with the aim of exploiting the complementary advantages of both: on the one hand, the stabilization of volatility and the improvement of risk-return efficiency guaranteed by BSC, on the other hand, the ability of DM to reduce exposure in regimes historically unfavorable to the market. This approach allows you to evaluate whether the integration of the two mechanisms can further improve performance compared to the standard model.



The analysis of equity lines shows how the combination of the two rules generates heterogeneous growth dynamics between the different strategies, reflecting the differentiated impact of the two filters on the time profile of returns. The *Enhanced* version continues to stand out for its significantly higher performance, the *Momentum-neutral* strategy worsens its performance, while the *Standard* and *Weakened* versions show overall limited performance.

Overall, the evolution of capital suggests that the integrated approach is able to effectively balance risk reduction with the ability to generate value over time, without distorting the underlying logic of strategies that combine *expected skewness* with *momentum*.

Comparison with wealth curves where the two rules are applied individually allows us to better understand the advantages of this combination.
The equity line associated with the Daniel and Moskowitz rule alone shows an overall more cautious trend, with lower volatility and more limited cumulative growth, a sign that reducing exposure in unfavorable regimes tends to sacrifice part of the potential return. On the contrary, the exclusive application of *volatility targeting* according to Barroso and Santa Clara produces more pronounced growth profiles, but exposes strategies to phases of greater volatility and drawdown. The sequential approach adopted in this analysis can be interpreted as a middle ground between these two extremes: on the one hand, it preserves the expansion capacity typical of BSC strategies, on the other hand, it benefits from the defensive component introduced by the DM filter, with a more balanced and robust overall result than both isolated implementations.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Vol	Skew	Q01
STD_DM_BSC	0,0560131	0,41704	0,70733	0,3579	0,16092	1,4054	-0,08564
WEAKENED_DM_BSC	0,033134	0,29035	0,42735	0,30933	0,15151	0,37639	-0,12384
ENHANCED_DM_BSC	0,13	0,86131	1,5898	0,20011	0,15669	0,716	-0,088961
MN_DM_BSC	0,079929	0,55798	0,92554	0,27869	0,16096	0,62412	-0,11737

Strategy	SortinoMod	Calmar	Robust	Omega
STD_DM_BSC	0,19621	0,1565	0,032038	-0,43203
WEAKENED_DM_BSC	0,1177	0,10712	0,0036025	-0,66359
ENHANCED_DM_BSC	0,4272	0,64964	0,36169	1,2225
MN_DM_BSC	0,27031	0,2868	0,091042	-0,024019
BENCH	0,25137	0,20476	0,20145	0

The analysis of performance and volatility metrics shows a more nuanced picture than is highlighted by equity lines alone.
While the combined approach confirms an overall improvement in volatility metrics, with a widespread reduction in maximum *drawdown*, and better stability of returns compared to single-rule versions, the increase in performance metrics is more limited. In particular, it is the *Enhanced DM+BSC* strategy that clearly stands out for its ability to generate

superior risk-adjusted returns, while the other versions show less significant improvements or values substantially like their counterparts where only the Barroso and Santa Clara rule is applied.

The picture becomes even clearer by looking at Omega metrics, which summarize the overall judgment on the quality of strategies. In this case, the improvement is broader and more widespread: all DM+BSC strategies record higher values than the versions with individual rules, with the exception of the *Momentum neutral* portfolio, which can be observed to deteriorate compared to its version to which only the *volatility targeting* rule is applied.
Particularly relevant, also in this case, is the result obtained by the *Enhanced DM+BSC* strategy, which reaches the highest value of Omega and is confirmed as the most effective solution in balancing return, risk and robustness, also generating the best return in the observation period.

Fama French regression with five factors:

The Fama–French five-factor regression is one of the most widely used tools in the empirical financial literature to analyze and decompose the performance of a portfolio or investment strategy. The idea behind this approach is to verify whether the observed returns can be explained by systematic exposures to a set of risk factors well documented in academic literature, or whether there is an unexplained return component, indicative of an ability to generate added value, i.e. alpha, through stock picking or market *timing*. The estimated model is as follows:

$$R_{p,t} - R_{f,t} = \alpha + \beta_{MKT}(R_{MKT,t} - R_{f,t}) + \beta_{SMB}SMB_t + \beta_{HML}HML_t + \beta_{RMW}RMW_t + \beta_{CMA}CMA_t + \varepsilon_t$$

Where $R_{p,t}$ is the return of the strategy or portfolio over the period t, $R_{f,t}$ is the risk-free rate, and the five factors to the right of the regression represent the main sources of systematic risk identified by Fama and French. The error ε_t term captures the yield component not explained by the model.

The analysis carried out through this regression analyzes several statistics:

- *Alpha*: indicates the average monthly excess return not explained by risk factors. A positive and meaningful value suggests that the strategy generates added value beyond what is justified by the risk taken.
- *Annualized Alpha*: Annualization of alpha, useful for comparing the excess return on an annual basis with that of other portfolios or benchmarks.
- *tAlpha*: measures the statistical significance of alpha; values above 1.96 mark the significance of this statistic.
- *Beta_MKT*: sensitivity of the portfolio to the market factor, high values indicate pro-cyclical behavior; market-neutral strategies close to zero*; negative a countercyclical profile.*
- *Beta_SMB*: exposure to the size factor, positive for small-cap preference, negative for large-cap.
- *Beta_HML*: reveals the inclination of the portfolio: positive if oriented towards *value investing* (i.e. there is a tendency to invest in more mature companies), negative if oriented towards *growth investing* (i.e. companies with high growth potential and high market multiples).

- *Beta_RMW*: Measures exposure to the profitability factor. A positive value indicates a preference for more profitable companies, while a negative value indicates a bias toward less profitable companies.
- *Beta_CMA*: measures exposure to the factor related to the company's investment policy. A positive value indicates a preference for companies with prudent investment strategies and gradual growth, while a negative value signals a bias toward companies that invest more aggressively and expansively.
- *R²*: Share of variance in returns explained by the model; high values (above 0.6) indicate that the factors describe the strategy well, low values (below 0.3) are typical of alternative or long-short strategies.

Strategia	Alpha	Alpha_ann	tAlpha	Beta_MKT	Beta_SMB	Beta_HML	Beta_RMW	Beta_CMA	R2
STD	0,0011383	0,013746	0,58375	-0,098994	-0,173	-0,37705	-0,22615	0,46351	0,22394
WEAKENED	-0,0013181	-0,015703	-0,71588	0,15943	0,095794	0,034933	-0,18184	-0,09756	0,14917
ENHANCED	0,0095808	0,12123	3,955	-0,45826	-0,29787	-0,57802	-0,056151	0,26022	0,50587
NEUTRAL	0,0101863	0,13843	3,5749	-0,62274	-0,36431	-0,63778	0,13714	0,38348	0,51496
STD_DM_BSC	0,0053875	0,066601	1,6935	-0,057641	-0,1912	-0,34198	-0,30558	0,33629	0,081218
WEAKENED_DM_BSC	0,0027958	0,034071	0,95373	0,14014	0,17413	-0,071629	-0,39322	-0,18349	0,11341
ENHANCED_DM_BSC	0,012338	0,15853	4,4665	-0,3787	-0,13901	-0,23231	0,1333	-0,10321	0,26629
NEUTRAL_DM_BSC	0,0084004	0,1056	2,9961	-0,41217	-0,20707	-0,072239	0,33849	-0,10025	0,28761

The results of the Fama–French regression are in line with the evidence shown by the *equity lines* and metrics for the respective strategies, albeit with some significant nuances. In the *backtest* period, the *Enhanced* and *Momentum neutral* base strategies show significantly higher cumulative growth than the *Standard* and *Weakened portfolios*, confirming the presence of a positive and statistically significant alpha. The two α-most basic strategies , in particular, show opposite logics and trends compared to the other two base portfolios. These two winning portfolios, in fact, have largely negative coefficients in terms of *value investing*, exposure to profitability factors and sensitivity to market returns, thus demonstrating that they are portfolios linked to countercyclical profiles with investment trends aimed at smaller companies with high growth potential.

The DM_BSC versions confirm and amplify the differences between the various strategies, showing a further increase in α for all portfolios except *Neutral*, showing consistency with wealth curves and metrics.
Specifically, the *Enhanced* portfolio reaches values close to 15.8% per year with high statistical significance. This behaviour highlights the combined effectiveness of Daniel

Moskowitz and Barroso Santa-Clara's rules in improving the risk-return profile through a controlled reduction of exposure in unfavourable market regimes and a general flattening of all factor ratios.

Looking at the fit coefficients, a clear gradient emerges between the different families of strategies: the higher values for *Enhanced* and *Neutral* show that a significant part of the variance in returns is still explained by the Fama–French factors, while the DM_BSC versions have much lower R^2, testifying to their more autonomous nature and poorly correlated to traditional market drivers. This reduction in the explanatory power of the model reflects precisely the desired effect of dynamic risk management rules: to contain systematic exposure without compromising the ability to generate alpha.
Overall, the regression and fit coefficients confirm that *Enhanced* and especially *Enhanced_DM_BSC* strategies represent the most efficient configurations, capable of producing superior and more stable returns through a progressive decorrelation from market risk factors.

Concluding remarks

The empirical analysis conducted largely confirms the evidence presented by Jacobs, Regele and Weber, demonstrating that the relationship between *expected skewness* and *momentum* continues to represent a significant dimension in explaining equity returns even in a modern market environment.
While operating over a more recent period and with a smaller sample size, the replication shows that the inclusion of expected *skewness* in the stock selection process allows for improved return quality and the overall stability of *momentum strategies*.

The evidence obtained follows the original logic: the Enhanced portfolio maintains, in all its versions, a clear superiority over all other configurations, confirming that the combination of high *momentum* and negative *expected skewness* does not aim to mechanically follow trends, but to favor stocks that express more regular positive returns and are less exposed to sudden crashes. In this way, the strategy retains the premium component of *momentum* but reduces its sensitivity to market *reversals*, improving its structural robustness. This approach makes it possible to preserve the premium component of momentum while mitigating its typical vulnerability to market reversal phases.

In a complementary way, the *Weakened strategy*, based on stocks with high positive *skewness* and high *momentum*, continues to show a weak and unstable profile, confirming the speculative and inefficient nature of these configurations. This contrast, also observed in the original paper, remains evident in the reply, testifying to the structural persistence of the link between asymmetry of returns and profitability of the *momentum factor*.

A further important element is represented by the introduction of dynamic risk management rules, the joint application of the Barroso-Santa Clara and Daniel-Moskowitz methodologies has made it possible to obtain the most solid and regular results of the entire experiment. In particular, the Enhanced DM+BSC strategy shows a clear overall improvement in the risk-return profile in the various strategies, with limited volatility, smaller *drawdowns* and more consistent returns over time. Dynamic leverage, adjusted for estimated volatility, and reduced exposure in adverse regimes have made it possible to optimise cumulative growth while preserving the stability of the wealth curve.

The most significant result emerges from the Omega metric, the *Enhanced* DM+BSC portfolio records a value of 1.222, corresponding to an average improvement of 122.2% compared to the benchmark, which by definition has an Omega value of zero.
This result highlights the ability of the combination of *expected skewness*, *momentum*

and dynamic risk management to produce more efficient, stable and resilient returns over time than the reference passive portfolio.

A methodological aspect that deserves attention, however, is the absence of transaction costs in the *backtest*, although this choice allows to isolate the purely theoretical effects of the relationship between *momentum* and *skewness*, it can amplify net returns and make the comparison with the operating world less realistic. In fact, the strategies analyzed imply a high monthly *turnover and a continuous* long-short *rebalancing*. By including realistic trading costs, risk-adjusted returns would be lower, while not altering the hierarchy between different setups.